

TCS NQT 2024 Previous Year Paper

Full-Length Mock Paper (Simulated) - 2024

TCS NQT Preparation Material
Generated on: April 05, 2026

General Instructions

- Total Duration: 120 minutes (No sectional time limit)
- Total Questions: 62 (20 QA + 15 LR + 15 VA + 10 PL + 2 Coding)
- Marking Scheme: +1 for correct MCQs, No negative marking
- Coding Section: Marks based on test cases passed
- No switching between sections once submitted
- Use of calculator is NOT permitted
- All questions are compulsory

Marking Scheme

Section	Questions	Marks/Question	Total Marks	Suggested Time
Quantitative Aptitude	20	+1	20	25 min
Logical Reasoning	15	+1	15	20 min
Verbal Ability	15	+1	15	15 min
Programming Logic	10	+1	10	15 min
Coding	2	Variable	~20-30	35-45 min

Note: There is NO negative marking. Coding questions are graded based on the number of test cases passed.

Section 1: Quantitative Aptitude

20 Questions | Topics: Number System, Percentages, Algebra, Geometry, Data Interpretation

Q1. [Medium]

If $x + \frac{1}{x} = 5$, find the value of $x^3 + \frac{1}{x^3}$.

- A) 110
- B) 125
- C) 140
- D) 95

Q2. [Easy]

A sum of Rs 8000 is invested at 10% p.a. compound interest. What is the amount after 3 years?

- A) Rs 10,648
- B) Rs 10,400
- C) Rs 11,000
- D) Rs 10,560

Q3. [Easy]

In a class, the average age of 30 boys is 14 years and the average age of 20 girls is 12 years. What is the average age of the whole class?

- A) 12.8 years
- B) 13.2 years
- C) 13.6 years
- D) 14.0 years

Q4. [Easy]

A train 150 m long passes a pole in 15 seconds. Find the speed of the train in km/h.

- A) 36 km/h
- B) 40 km/h
- C) 45 km/h
- D) 30 km/h

Q5. [Easy]

If 40% of a number is added to 42, the result is the number itself. Find the number.

- A) 60**
- B) 70**
- C) 80**
- D) 90**

Q6. [Easy]

The ratio of the ages of A and B is 3:5. If the sum of their ages is 64, what will be A's age after 8 years?

- A) 32**
- B) 30**
- C) 34**
- D) 28**

Q7. [Medium]

A shopkeeper marks the price of an article 30% above the cost price and gives a discount of 10%. Find his profit percentage.

- A) 15%**
- B) 17%**
- C) 20%**
- D) 13%**

Q8. [Medium]

Two pipes A and B can fill a tank in 12 hours and 18 hours respectively. If both are opened together, how long to fill the tank?

- A) 7.2 hours**
- B) 6.5 hours**
- C) 8 hours**
- D) 7.5 hours**

Q9. [Easy]

In how many ways can 5 people be seated in a row?

- A) 60**
- B) 120**
- C) 24**
- D) 720**

Q10. [Medium]

A boat goes 24 km upstream in 6 hours and 24 km downstream in 4 hours. Find the speed of the boat in still water.

- A)** 5 km/h
- B)** 6 km/h
- C)** 4 km/h
- D)** 7 km/h

Q11. [Easy]

The LCM of two numbers is 120 and their HCF is 10. If one number is 40, find the other.

- A)** 20
- B)** 30
- C)** 40
- D)** 60

Q12. [Easy]

A man buys an article for Rs 450 and sells it at a loss of 20%. What is the selling price?

- A)** Rs 350
- B)** Rs 360
- C)** Rs 380
- D)** Rs 400

Q13. [Easy]

If the perimeter of a rectangle is 56 cm and the length is 18 cm, find the breadth.

- A)** 8 cm
- B)** 10 cm
- C)** 12 cm
- D)** 14 cm

Q14. [Medium]

Study the bar chart: Company X's sales (in lakhs) for 2019-2023 are 30, 45, 35, 55, 60. In which year was the percentage increase in sales the highest compared to the previous year?

- A)** 2020
- B)** 2021
- C)** 2022
- D)** 2023

Q15. [Hard]

A cistern has a leak which can empty it in 8 hours. A tap is opened which admits 6 litres per hour into the cistern, and it is now emptied in 12 hours. How many litres does the cistern hold?

- A) 120**
- B) 144**
- C) 100**
- D) 150**

Q16. [Easy]

The diagonal of a square is $10\sqrt{2}$ cm. Find the area of the square.

- A) 100 sq cm**
- B) 50 sq cm**
- C) 200 sq cm**
- D) 150 sq cm**

Q17. [Medium]

In a mixture of 60 litres, the ratio of milk to water is 2:1. How much water must be added to make the ratio 1:2?

- A) 40 litres**
- B) 50 litres**
- C) 60 litres**
- D) 30 litres**

Q18. [Easy]

The average of first 50 natural numbers is:

- A) 25**
- B) 25.5**
- C) 26**
- D) 24.5**

Q19. [Hard]

A number when divided by 342 gives a remainder 47. When the same number is divided by 19, what will be the remainder?

- A) 9**
- B) 7**
- C) 3**
- D) 5**

Q20. [Easy]

The simple interest on Rs 5000 at a certain rate for 2 years is Rs 800. Find the rate.

- A) 6%**
- B) 8%**
- C) 10%**
- D) 12%**

Section 2: Logical Reasoning

15 Questions | Topics: Series, Coding-Decoding, Arrangements, Puzzles, Blood Relations

Q1. [Easy]

Find the next number in the series: 2, 6, 12, 20, 30, ?

- A)** 40
- B)** 42
- C)** 44
- D)** 38

Q2. [Easy]

In a certain code language, COMPUTER is written as DPNQVUFS. How is PROGRAM written?

- A)** QSPHSBN
- B)** QSPHSBO
- C)** QSPHSAN
- D)** QRPHSBN

Q3. [Medium]

A is the mother of B. B is the sister of C. D is the son of C. How is A related to D?

- A)** Mother
- B)** Grandmother
- C)** Aunt
- D)** Cannot be determined

Q4. [Hard]

Five friends P, Q, R, S, T are sitting in a row facing north. R is to the immediate right of P. T is at one of the ends. Q is between S and R. Who is in the middle?

- A)** P
- B)** Q
- C)** R
- D)** S

Q5. [Easy]

If FRIEND is coded as 6-18-9-5-14-4, how is CANDLE coded?

- A) 3-1-14-4-12-5**
- B) 3-2-14-4-12-5**
- C) 3-1-13-4-12-5**
- D) 4-1-14-4-12-5**

Q6. [Easy]

Pointing to a photograph, a man says, 'The boy in the photo is the son of the only son of my mother.' How is the boy related to the man?

- A) Brother**
- B) Son**
- C) Nephew**
- D) Uncle**

Q7. [Medium]

Statement: All books are pens. Some pens are pencils. Conclusion I: Some books are pencils. Conclusion II: Some pencils are books.

- A) Only I follows**
- B) Only II follows**
- C) Both follow**
- D) Neither follows**

Q8. [Medium]

Which number replaces the question mark? 3, 7, 15, 31, 63, ?

- A) 115**
- B) 121**
- C) 127**
- D) 131**

Q9. [Easy]

Arrange the following in meaningful order: 1.Key, 2.Door, 3.Lock, 4.Room, 5.Switch on light

- A) 1,3,2,4,5**
- B) 3,1,2,5,4**
- C) 1,2,3,4,5**
- D) 1,3,2,5,4**

Q10. [Easy]

In a row of students, Ram is 15th from the left and 20th from the right. How many students are in the row?

- A)** 33
- B)** 34
- C)** 35
- D)** 36

Q11. [Medium]

If 'ORANGE' is coded as 'NAQDFD', how is 'BANANA' coded?

- A)** AZMZMZ
- B)** CBOBOB
- C)** AZMBMZ
- D)** AZMZMB

Q12. [Medium]

A clock shows 3:15. What is the angle between the hour and minute hands?

- A)** 0 degrees
- B)** 7.5 degrees
- C)** 15 degrees
- D)** 22.5 degrees

Q13. [Medium]

If South-East becomes North, what does North-West become?

- A)** South
- B)** East
- C)** North-East
- D)** South-West

Q14. [Easy]

Study the Venn diagram: 40 students play cricket, 30 play football, 15 play both. How many play only cricket?

- A)** 25
- B)** 30
- C)** 15
- D)** 40

Q15. [Easy]

Complete the pattern: AZ, BY, CX, DW, ?

- A)** EV
- B)** EU
- C)** FV
- D)** EW

Section 3: Verbal Ability

15 Questions | Topics: Reading Comprehension, Grammar, Vocabulary, Sentence Correction

Q1. [Easy]

Read the passage: 'Artificial intelligence is transforming industries across the globe. From healthcare to finance, AI-driven solutions are increasing efficiency and reducing human error. However, concerns about job displacement and ethical implications remain.' What is the primary theme of the passage?

- A)** AI will replace all jobs
- B)** AI's impact on industries with associated concerns
- C)** AI is only useful in healthcare
- D)** Ethical issues make AI unusable

Q2. [Easy]

Choose the word most similar in meaning to 'UBIQUITOUS':

- A)** Rare
- B)** Omnipresent
- C)** Hidden
- D)** Unique

Q3. [Easy]

Identify the error: 'Each of the students have completed their assignment on time.'

- A)** Each of
- B)** have completed
- C)** their assignment
- D)** on time

Q4. [Easy]

Choose the correct sentence:

- A)** He don't know nothing about it.
- B)** He doesn't know anything about it.
- C)** He don't know anything about it.
- D)** He doesn't know nothing about it.

Q5. [Medium]

Fill in the blank: 'The manager, along with his team members, ____ present at the meeting.'

- A) were**
- B) was**
- C) are**
- D) have been**

Q6. [Easy]

Choose the word opposite in meaning to 'VERBOSE':

- A) Talkative**
- B) Concise**
- C) Lengthy**
- D) Detailed**

Q7. [Easy]

The idiom 'Burning the midnight oil' means:

- A) Wasting resources**
- B) Working late into the night**
- C) Starting a fire**
- D) Getting angry**

Q8. [Medium]

Rearrange to form a meaningful sentence: (P) the company (Q) despite financial challenges (R) managed to (S) launch its new product

- A) P-Q-R-S**
- B) P-R-S-Q**
- C) Q-P-R-S**
- D) P-S-Q-R**

Q9. [Easy]

Choose the correctly spelled word:

- A) Accomodation**
- B) Accommodation**
- C) Acommodation**
- D) Acomodation**

Q10. [Easy]

Read the passage: 'Climate change is one of the most pressing issues of our time. Rising sea levels, extreme weather events, and loss of biodiversity threaten ecosystems worldwide. International cooperation is essential to mitigate these effects.' According to the passage, what is essential to address climate change?

- A) Individual action only**
- B) International cooperation**
- C) Technological innovation**
- D) Ignoring the problem**

Q11. [Easy]

Choose the correct preposition: 'She is proficient ____ several programming languages.'

- A) at**
- B) in**
- C) with**
- D) on**

Q12. [Easy]

Select the word that best completes the analogy: Book : Author :: Painting : ?

- A) Canvas**
- B) Brush**
- C) Artist**
- D) Gallery**

Q13. [Easy]

Identify the type of sentence: 'What a beautiful sunset!'

- A) Declarative**
- B) Interrogative**
- C) Imperative**
- D) Exclamatory**

Q14. [Medium]

Choose the word most similar to 'PRAGMATIC':

- A) Idealistic**
- B) Practical**
- C) Theoretical**
- D) Abstract**

Q15. [Medium]

Fill in the blank: 'Neither the teacher nor the students ____ aware of the schedule change.'

- A)** was
- B)** were
- C)** is
- D)** has been

Section 4: Programming Logic

10 Questions | Topics: Output Prediction, Code Snippets, DSA MCQs

Q1. [Medium]

What is the output of the following C code?

```
#include <stdio.h>
int main() {
    int a = 5, b = 10;
    printf("%d", a+++b);
    return 0;
}
```

- A) 15
- B) 16
- C) Error
- D) Undefined

Q2. [Easy]

What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(\log n)$
- D) $O(1)$

Q3. [Easy]

What is the output?

```
x = [1, 2, 3]
y = x
y.append(4)
print(len(x))
```

- A) 3
- B) 4
- C) Error
- D) None

Q4. [Easy]

Which data structure uses LIFO principle?

- A) Queue**
- B) Stack**
- C) Linked List**
- D) Tree**

Q5. [Medium]

What is the output?

```
def f(x, lst=[]):  
    lst.append(x)  
    return lst  
print(f(1))  
print(f(2))
```

- A) [1] then [2]**
- B) [1] then [1, 2]**
- C) Error**
- D) [1, 2] then [1, 2]**

Q6. [Easy]

What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$**
- B) $O(n^2)$**
- C) $O(n)$**
- D) $O(\log n)$**

Q7. [Medium]

What does the following function compute?

```
int mystery(int n) {  
    if (n == 0) return 0;  
    return n % 10 + mystery(n / 10);  
}
```

- A) Reverse of n**
- B) Sum of digits of n**
- C) Number of digits in n**
- D) Factorial of n**

Q8. [Easy]

In a singly linked list, what is the time complexity to insert a node at the beginning?

- A)** $O(1)$
- B)** $O(n)$
- C)** $O(\log n)$
- D)** $O(n^2)$

Q9. [Easy]

What is the output?

```
#include <stdio.h>
int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int *ptr = arr + 3;
    printf("%d", *ptr);
}
```

- A)** 10
- B)** 30
- C)** 40
- D)** 50

Q10. [Easy]

Which of the following is NOT a valid sorting algorithm?

- A)** Radix Sort
- B)** Counting Sort
- C)** Random Sort
- D)** Shell Sort

Section 5: Coding Questions

2 Questions | One Easy + One Medium

Coding Question 1: Two Sum [Easy]

Problem Statement: Given an array of integers `nums` and an integer `target`, return the indices of the two numbers such that they add up to the target. You may assume that each input has exactly one solution, and you may not use the same element twice.

Input Format:

```
First line: n (size of array)
Second line: n space-separated integers
Third line: target value
```

Output Format:

```
Two space-separated indices (0-based) in ascending order
```

Example 1:

Input:

```
4
2 7 11 15
9
```

Output:

```
0 1
```

Explanation: $nums[0] + nums[1] = 2 + 7 = 9$

Example 2:

Input:

```
3
3 2 4
6
```

Output:

```
1 2
```

Explanation: $nums[1] + nums[2] = 2 + 4 = 6$

Constraints:

```
2 <= n <= 10^4
-10^9 <= nums[i] <= 10^9
Exactly one valid answer exists.
```

Coding Question 2: Longest Substring Without Repeating Characters [Medium]

Problem Statement: Given a string *s*, find the length of the longest substring without repeating characters.

Input Format:

A single string *s*

Output Format:

An integer representing the length of the longest substring without repeating characters

Example 1:

Input:

abcabcbb

Output:

3

Explanation: The answer is 'abc' with length 3.

Example 2:

Input:

bbbbbb

Output:

1

Explanation: The answer is 'b' with length 1.

Constraints:

$0 \leq s.length \leq 5 * 10^4$
s consists of English letters, digits, symbols and spaces.

Answer Key

Section 1: Quantitative Aptitude

Q#	Answer	Difficulty
1	A) 110	Medium
2	A) Rs 10,648	Easy
3	B) 13.2 years	Easy
4	A) 36 km/h	Easy
5	B) 70	Easy
6	A) 32	Easy
7	B) 17%	Medium
8	A) 7.2 hours	Medium
9	B) 120	Easy
10	A) 5 km/h	Medium
11	B) 30	Easy
12	B) Rs 360	Easy
13	B) 10 cm	Easy
14	C) 2022	Medium
15	B) 144	Hard
16	A) 100 sq cm	Easy
17	C) 60 litres	Medium
18	B) 25.5	Easy
19	A) 9	Hard
20	B) 8%	Easy

Section 2: Logical Reasoning

Q#	Answer	Difficulty
1	B) 42	Easy
2	A) QSPHSBN	Easy
3	B) Grandmother	Medium
4	C) R	Hard
5	A) 3-1-14-4-12-5	Easy
6	B) Son	Easy
7	D) Neither follows	Medium
8	C) 127	Medium

9	A) 1,3,2,4,5	Easy
10	B) 34	Easy
11	A) AZMZMZ	Medium
12	B) 7.5 degrees	Medium
13	A) South	Medium
14	A) 25	Easy
15	A) EV	Easy

Section 3: Verbal Ability

Q#	Answer	Difficulty
1	B) AI's impact on industries with associated concerns	Easy
2	B) Omnipresent	Easy
3	B) have completed	Easy
4	B) He doesn't know anything about it.	Easy
5	B) was	Medium
6	B) Concise	Easy
7	B) Working late into the night	Easy
8	B) P-R-S-Q	Medium
9	B) Accommodation	Easy
10	B) International cooperation	Easy
11	B) in	Easy
12	C) Artist	Easy
13	D) Exclamatory	Easy
14	B) Practical	Medium
15	B) were	Medium

Section 4: Programming Logic

Q#	Answer	Difficulty
1	A) 15	Medium
2	C) $O(\log n)$	Easy
3	B) 4	Easy
4	B) Stack	Easy
5	B) [1] then [1, 2]	Medium
6	B) $O(n^2)$	Easy

7	B) Sum of digits of n	Medium
8	A) $O(1)$	Easy
9	C) 40	Easy
10	C) Random Sort	Easy

Section 5: Coding Solutions

Q1 (Two Sum): Use a hash map to store each number and its index. For each number, check if (target - number) exists in the map. Time: $O(n)$, Space: $O(n)$.

Q2 (Longest Substring Without Repeating Characters): Use the sliding window technique with a set/map. Maintain a window [left, right] and expand right. If a duplicate is found, shrink from left. Track the maximum window size. Time: $O(n)$, Space: $O(\min(m, n))$ where m is the character set size.

Detailed Solutions

Section 1: Quantitative Aptitude

Q1. [Medium]

If $x + 1/x = 5$, find the value of $x^3 + 1/x^3$.

Answer: A) 110

Explanation: $x + 1/x = 5$. Cubing both sides using the identity $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$: $x^3 + 1/x^3 + 3 \cdot 5 = 125$, so $x^3 + 1/x^3 = 125 - 15 = 110$.

Q2. [Easy]

A sum of Rs 8000 is invested at 10% p.a. compound interest. What is the amount a...

Answer: A) Rs 10,648

Explanation: $A = P(1+r/100)^n = 8000 \cdot (1.1)^3 = 8000 \cdot 1.331 = 10648$.

Q3. [Easy]

In a class, the average age of 30 boys is 14 years and the average age of 20 gir...

Answer: B) 13.2 years

Explanation: Total age = $30 \cdot 14 + 20 \cdot 12 = 420 + 240 = 660$. Average = $660/50 = 13.2$ years.

Q4. [Easy]

A train 150 m long passes a pole in 15 seconds. Find the speed of the train in k...

Answer: A) 36 km/h

Explanation: Speed = $150/15 = 10$ m/s = $10 \cdot 18/5 = 36$ km/h.

Q5. [Easy]

If 40% of a number is added to 42, the result is the number itself. Find the num...

Answer: B) 70

Explanation: Let number = x . $0.4x + 42 = x \Rightarrow 0.6x = 42 \Rightarrow x = 70$.

Q6. [Easy]

The ratio of the ages of A and B is 3:5. If the sum of their ages is 64, what wi...

Answer: A) 32

Explanation: $A = 3 \times 64 / 8 = 24$, $B = 40$. A's age after 8 years = $24 + 8 = 32$.

Q7. [Medium]

A shopkeeper marks the price of an article 30% above the cost price and gives a ...

Answer: B) 17%

Explanation: Let $CP=100$. $MP=130$. $SP=130 \times 0.9=117$. Profit% = 17%.

Q8. [Medium]

Two pipes A and B can fill a tank in 12 hours and 18 hours respectively. If both...

Answer: A) 7.2 hours

Explanation: Combined rate = $1/12 + 1/18 = (3+2)/36 = 5/36$. Time = $36/5 = 7.2$ hours.

Q9. [Easy]

In how many ways can 5 people be seated in a row?

Answer: B) 120

Explanation: $5! = 120$.

Q10. [Medium]

A boat goes 24 km upstream in 6 hours and 24 km downstream in 4 hours. Find the ...

Answer: A) 5 km/h

Explanation: Upstream speed = 4 km/h, downstream speed = 6 km/h. Speed in still water = $(4+6)/2 = 5$ km/h.

Q11. [Easy]

The LCM of two numbers is 120 and their HCF is 10. If one number is 40, find the...

Answer: B) 30

Explanation: $LCM \times HCF = \text{product of numbers}$. $120 \times 10 = 40 \times x \Rightarrow x = 30$.

Q12. [Easy]

A man buys an article for Rs 450 and sells it at a loss of 20%. What is the sell...

Answer: B) Rs 360

Explanation: $SP = 450 * 0.8 = 360$.

Q13. [Easy]

If the perimeter of a rectangle is 56 cm and the length is 18 cm, find the bread...

Answer: B) 10 cm

Explanation: $2(l+b) = 56 \Rightarrow l+b = 28 \Rightarrow b = 28-18 = 10$ cm.

Q14. [Medium]

Study the bar chart: Company X's sales (in lakhs) for 2019-2023 are 30, 45, 35, ...

Answer: C) 2022

Explanation: Increases: 2020: 50%, 2021: -22.2%, 2022: 57.1%, 2023: 9.1%. Highest is 2022 at 57.1%.

Q15. [Hard]

A cistern has a leak which can empty it in 8 hours. A tap is opened which admits...

Answer: B) 144

Explanation: Rate of leak = $C/8$. With tap, $(C/8 - 6)$ per hour empties in 12 hours. So $C = 12*(C/8-6) \Rightarrow C = 1.5C-72 \Rightarrow 0.5C = 72 \Rightarrow C = 144$.

Q16. [Easy]

The diagonal of a square is $10\sqrt{2}$ cm. Find the area of the square.

Answer: A) 100 sq cm

Explanation: Diagonal = $a\sqrt{2} = 10\sqrt{2}$, so $a = 10$. Area = 100 sq cm.

Q17. [Medium]

In a mixture of 60 litres, the ratio of milk to water is 2:1. How much water must...

Answer: C) 60 litres

Explanation: Milk = 40, Water = 20. New ratio 1:2 means water = 80. Water to add = 80-20 = 60.

Q18. [Easy]

The average of first 50 natural numbers is:

Answer: B) 25.5

Explanation: Sum = $50 \times 51 / 2 = 1275$. Average = $1275 / 50 = 25.5$.

Q19. [Hard]

A number when divided by 342 gives a remainder 47. When the same number is divided...

Answer: A) 9

Explanation: $N = 342k + 47 = 19 \times 18k + 19 \times 2 + 9 = 19(18k+2) + 9$. Remainder = 9.

Q20. [Easy]

The simple interest on Rs 5000 at a certain rate for 2 years is Rs 800. Find the...

Answer: B) 8%

Explanation: $SI = \frac{PRT}{100} \Rightarrow 800 = \frac{5000 \times R \times 2}{100} \Rightarrow R = 8\%$.

Section 2: Logical Reasoning

Q21. [Easy]

Find the next number in the series: 2, 6, 12, 20, 30, ?

Answer: B) 42

Explanation: Differences: 4, 6, 8, 10, 12. Pattern: $n \times (n+1)$. Next = $6 \times 7 = 42$.

Q22. [Easy]

In a certain code language, COMPUTER is written as DPNQVUFS. How is PROGRAM writ...

Answer: A) QSPHSBN

Explanation: Each letter is shifted by +1. P->Q, R->S, O->P, G->H, R->S, A->B, M->N.

Q23. [Medium]

A is the mother of B. B is the sister of C. D is the son of C. How is A related ...

Answer: B) Grandmother

Explanation: A is mother of B. B is sister of C, so A is mother of C. D is son of C. So A is grandmother of D.

Q24. [Hard]

Five friends P, Q, R, S, T are sitting in a row facing north. R is to the immedi...

Answer: C) R

Explanation: Arrangement: T, S, Q, R, P or P, R, Q, S, T. Wait -- R is immediate right of P => ...P R... Q is between S and R => S Q R. So: T S Q R P. But T is at end -- this gives T, S, Q, R, P with R in middle? No, middle is Q (position 3). Actually rearranging: T, P, R, Q, S -- R immediate right of P, Q between R and S, T at end. Middle = R.

Q25. [Easy]

If FRIEND is coded as 6-18-9-5-14-4, how is CANDLE coded?

Answer: A) 3-1-14-4-12-5

Explanation: Each letter is replaced by its position: C=3, A=1, N=14, D=4, L=12, E=5.

Q26. [Easy]

Pointing to a photograph, a man says, 'The boy in the photo is the son of the on...

Answer: B) Son

Explanation: Only son of my mother = the man himself. So the boy is the man's son.

Q27. [Medium]

Statement: All books are pens. Some pens are pencils. **Conclusion I:** Some books a...

Answer: D) Neither follows

Explanation: All books are pens, some pens are pencils. The overlap of pens that are pencils may not include books. Neither conclusion necessarily follows.

Q28. [Medium]

Which number replaces the question mark? 3, 7, 15, 31, 63, ?

Answer: C) 127

Explanation: Pattern: $2^n - 1$. $3=4-1$, $7=8-1$, $15=16-1$, $31=32-1$, $63=64-1$, next = $128-1 = 127$.

Q29. [Easy]

Arrange the following in meaningful order: 1.Key, 2.Door, 3.Lock, 4.Room, 5.Swit...

Answer: A) 1,3,2,4,5

Explanation: Logical order: Key -> Lock -> Door -> Room -> Switch on light.

Q30. [Easy]

In a row of students, Ram is 15th from the left and 20th from the right. How man...

Answer: B) 34

Explanation: Total = $15 + 20 - 1 = 34$.

Q31. [Medium]

If 'ORANGE' is coded as 'NAQDFD', how is 'BANANA' coded?

Answer: A) AZMZMZ

Explanation: Each letter shifts by -1: O->N, R->Q, A->Z? Actually O->N (shifted -1), R->A? Let me re-check: ORANGE->NAQDFD. O(-1)=N, R(-1)=Q, A(-1)=Z? No. Looking again: O=15->N=14(-1), R=18->A=1? That's not -1. The pattern is reversing position: each letter mapped to reverse alphabet (A=Z, B=Y, etc). O->L? No. Actually encoding is -1 for each: O->N, R->Q, A->Z(wrap), N->M, G->F, E->D. So BANANA: B->A, A->Z, N->M, A->Z, N->M, A->Z = AZMBMZ? But answer choice A is AZMBMZ... wait let me recount. BANANA: B(-1)=A, A(-1)=Z, N(-1)=M, A(-1)=Z, N(-1)=M, A(-1)=Z => AZMZMZ. Yes.

Q32. [Medium]

A clock shows 3:15. What is the angle between the hour and minute hands?

Answer: B) 7.5 degrees

Explanation: At 3:15, minute hand at 90 degrees (pointing at 3). Hour hand at $90 + 15 \times 0.5 = 97.5$ degrees. Angle = 7.5 degrees.

Q33. [Medium]

If South-East becomes North, what does North-West become?

Answer: A) South

Explanation: SE $\rightarrow$ N is a rotation of 135 degrees clockwise. Applying same rotation to NW: NW + 135 CW = S.

Q34. [Easy]

Study the Venn diagram: 40 students play cricket, 30 play football, 15 play both...

Answer: A) 25

Explanation: Only cricket = $40 - 15 = 25$.

Q35. [Easy]

Complete the pattern: AZ, BY, CX, DW, ?

Answer: A) EV

Explanation: First letter A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ E. Second letter Z $\rightarrow$ Y $\rightarrow$ X $\rightarrow$ W $\rightarrow$ V. Answer: EV.

Section 3: Verbal Ability

Q36. [Easy]

Read the passage: 'Artificial intelligence is transforming industries across the...

Answer: B) AI's impact on industries with associated concerns

Explanation: The passage discusses both the positive impact and the concerns, making B the best answer.

Q37. [Easy]

Choose the word most similar in meaning to 'UBIQUITOUS':

Answer: B) Omnipresent

Explanation: Ubiquitous means present everywhere, which is synonymous with omnipresent.

Q38. [Easy]

Identify the error: 'Each of the students have completed their assignment on tim...

Answer: B) have completed

Explanation: 'Each' is singular and takes 'has completed' not 'have completed'.

Q39. [Easy]

Choose the correct sentence:

Answer: B) He doesn't know anything about it.

Explanation: Correct subject-verb agreement (doesn't with he) and avoids double negative.

Q40. [Medium]

Fill in the blank: 'The manager, along with his team members, ____ present at the...

Answer: B) was

Explanation: 'Along with' does not change the subject. 'Manager' is singular, so 'was' is correct.

Q41. [Easy]

Choose the word opposite in meaning to 'VERBOSE':

Answer: B) Concise

Explanation: Verbose means using too many words; its antonym is concise.

Q42. [Easy]

The idiom 'Burning the midnight oil' means:

Answer: B) Working late into the night

Explanation: This idiom means studying or working late at night.

Q43. [Medium]

Rearrange to form a meaningful sentence: (P) the company (Q) despite financial c...

Answer: B) P-R-S-Q

Explanation: The company managed to launch its new product despite financial challenges.

Q44. [Easy]

Choose the correctly spelled word:

Answer: B) Accommodation

Explanation: Accommodation has double 'c' and double 'm'.

Q45. [Easy]

Read the passage: 'Climate change is one of the most pressing issues of our time...

Answer: B) International cooperation

Explanation: The passage explicitly states international cooperation is essential.

Q46. [Easy]

Choose the correct preposition: 'She is proficient ____ several programming langu...

Answer: B) in

Explanation: 'Proficient in' is the correct collocation.

Q47. [Easy]

Select the word that best completes the analogy: Book : Author :: Painting : ?

Answer: C) Artist

Explanation: A book is created by an author; a painting is created by an artist.

Q48. [Easy]

Identify the type of sentence: 'What a beautiful sunset!'

Answer: D) Exclamatory

Explanation: It expresses strong emotion and ends with an exclamation mark.

Q49. [Medium]

Choose the word most similar to 'PRAGMATIC':

Answer: B) Practical

Explanation: Pragmatic means dealing with things in a practical way.

Q50. [Medium]

Fill in the blank: 'Neither the teacher nor the students ____ aware of the schedu...

Answer: B) were

Explanation: With 'neither...nor', the verb agrees with the nearer subject ('students' - plural), so 'were'.

Section 4: Programming Logic

Q51. [Medium]

What is the output of the following C code?

```
#include <stdio.h><br...
```

Answer: A) 15

Explanation: $a+++b$ is parsed as $(a++) + b = 5 + 10 = 15$. After this, a becomes 6.

Q52. [Easy]

What is the time complexity of binary search on a sorted array of n elements?

Answer: C) $O(\log n)$

Explanation: Binary search halves the search space each time, giving $O(\log n)$ complexity.

Q53. [Easy]

What is the output?

```
x = [1, 2, 3]
y = x
y.append(4)
print(l...
```

Answer: B) 4

Explanation: $y = x$ creates a reference, not a copy. Both point to the same list. After `append`, x also has 4 elements.

Q54. [Easy]

Which data structure uses LIFO principle?

Answer: B) Stack

Explanation: Stack follows Last In First Out (LIFO) principle.

Q55. [Medium]

What is the output?

```
def f(x, lst=[]):  
    lst.append(x)<br/...
```

Answer: B) [1] then [1, 2]

Explanation: Default mutable arguments in Python are shared across calls. The list persists between calls.

Q56. [Easy]

What is the worst-case time complexity of QuickSort?

Answer: B) $O(n^2)$

Explanation: QuickSort's worst case (already sorted array with bad pivot) is $O(n^2)$.

Q57. [Medium]

What does the following function compute?

```
int mystery(int n) {  
    &nbs...;
```

Answer: B) Sum of digits of n

Explanation: It adds the last digit ($n\%10$) and recurses on the remaining digits ($n/10$).

Q58. [Easy]

In a singly linked list, what is the time complexity to insert a node at the beg...

Answer: A) $O(1)$

Explanation: Inserting at the head of a singly linked list only requires updating the head pointer - $O(1)$.

Q59. [Easy]

What is the output?

```
#include <stdio.h>  
int main() {  
    &nbsp;   ...;
```

Answer: C) 40

Explanation: arr + 3 points to the 4th element (index 3), which is 40.

Q60. [Easy]

Which of the following is NOT a valid sorting algorithm?

Answer: C) Random Sort

Explanation: Random Sort is not a standard sorting algorithm. Radix, Counting, and Shell Sort are all well-known algorithms.

Section 5: Coding Solutions (Detailed)

Q61 - Two Sum [Easy]

Approach: Use a hash map to store each number and its index. For each number, check if (target - number) exists in the map. Time: $O(n)$, Space: $O(n)$.

Q62 - Longest Substring Without Repeating Characters [Medium]

Approach: Use the sliding window technique with a set/map. Maintain a window [left, right] and expand right. If a duplicate is found, shrink from left. Track the maximum window size. Time: $O(n)$, Space: $O(\min(m, n))$ where m is the character set size.